

Individual Assignment 5

Natalie Chen

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Set up

Packages

```
#general use
library(here)

library(tidyverse)

library(janitor)

library(snakecase)

# packages for visualization customization

library(scales)
library(paletteer)
library(ggthemes)

# reading in .xlsx files
library(readxl)

library(NatParksPalettes)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

copied from class

```
# creating new object from sites
ncos_sites <- sites |>

  # filter to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

water_quality_clean <- water_quality |>

  # cleaning column names
  clean_names() |>

  # filtering join: only include sites in ncos_sites
  # keep all the rows on the left that match up with rows on the right
  semi_join(ncos_sites, by = "site") |>

  # combine date and site together and remove those indiv. columns
  unite("date_site", date, site, remove = TRUE) |>

  # group by date_site
  group_by(date_site) |>
```

```

# create new columns that show median of each variable at each date_site
# because there are multiple samples at the same site at different depths
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

# ungroup
ungroup() |>

# filter out the salinity outlier
filter(date_site != "2022-08-12_NVBR")

```

```

# creating new object from taxon_list
taxon_list_clean <- taxon_list |>

# converted taxon name values to snake case (lowercase)
mutate(verbatim_name = to_any_case(verbatim_name,
                                   case = "snake"))

```

```

# creating new clean aquatic invertebrates object from aquatic invertebrates
aq_ins_clean <- aq_ins |>

# cleaning column names
clean_names() |>

# filtering join: only include quarterly NCOS surveys from ncos_sites
semi_join(ncos_sites, by = "site") |>

# renaming column to just date
rename(date = date_on_vial) |>

# select columns of interest
select(date, sample_type, site,

       # 9 most abundant taxon
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,

```

```

    annelida_oligochaete,
    diptera_chironomid,
    annelida_polychaete) |>

# filter to only include filtered beaker samples
filter(sample_type %in% c("FB250", "FB 250")) |>

pivot_longer(
  # cols = columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>

# group by the date, site, and taxon
group_by(date, site, taxon) |>

# calculate average abundance per liter (sum abundance divided by 7.5
  ↳ liters)
summarize(avg_per_lit = sum(abundance, na.rm = TRUE)/7.5) |>

# ungroup
ungroup() |>

# create a new column called `date_site` to join with water quality
# don't remove the original columns
unite("date_site", date, site, remove = FALSE) |>

# join with water quality clean
# wqc has `date_site` column as well
left_join(water_quality_clean, by = "date_site") |>

# join with taxonomic information from taxon list
# verbatim_name column in taxon_list_clean matches with taxon column in
  ↳ aq_ins_clean
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",

```

```

site == "NEC" ~ "East Channel",
site == "NMC" ~ "Main Channel",
site == "NPB" ~ "South of Phelps Bridge",
site == "NPB1" ~ "North of Phelps Bridge",
site == "NPB2" ~ "Phelps Road",
site == "NWP" ~ "West Pond",
site == "NDC" ~ "Devereux Creek"
)) |>

# setting factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
↪   = TRUE),
       site_full = fct_rev(site_full))

```

```

salinity_plot <- ggplot(data = aq_ins_clean,
                        # x-axis: full site name
                        # y-axis: median salinity
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
# points representing median salinity at each survey
geom_jitter(height = 0,
            shape = 21) +

stat_summary(geom = "point",
            fun = median,
            color = "red",
            size = 2,) +

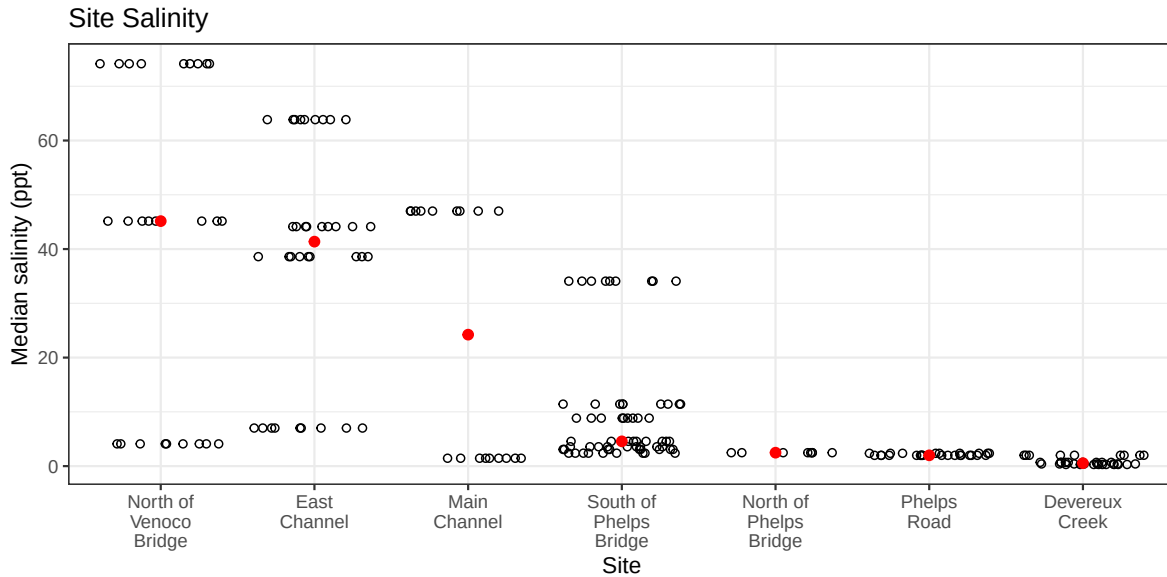
# wrap x-axis labels
scale_x_discrete(labels = label_wrap(width = 10)) +

# labeling titles and axes
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site Salinity") +

# change theme
theme_bw()

# displaying the object
salinity_plot

```



```
copepods <- aq_ins_clean |>
```

```
  filter(taxon == "copepod") |>
```

```
  mutate(log_avg_per_lit = log(avg_per_lit + 1))
```

```
# creating object to store plot
```

```
copepod_plot <- ggplot(data = copepods,
  # x-axis: full site name
  # y-axis: log transformed avg count/L
  mapping = aes(x = site_full,
    y = log_avg_per_lit)) +
```

```
  # jitter to represent salinity
  geom_jitter(height = 0,
    width = 0.1,
    shape = 21) +
```

```
# points to represent median copepod abundance
```

```
  stat_summary(geom = "point",
    fun = median,
    color = "blue",
    size = 2) +
```

```
  # wrapping x axis labels
```

```
  scale_x_discrete(labels = label_wrap(width = 10)) +
```

```
  # relabelling axes
```

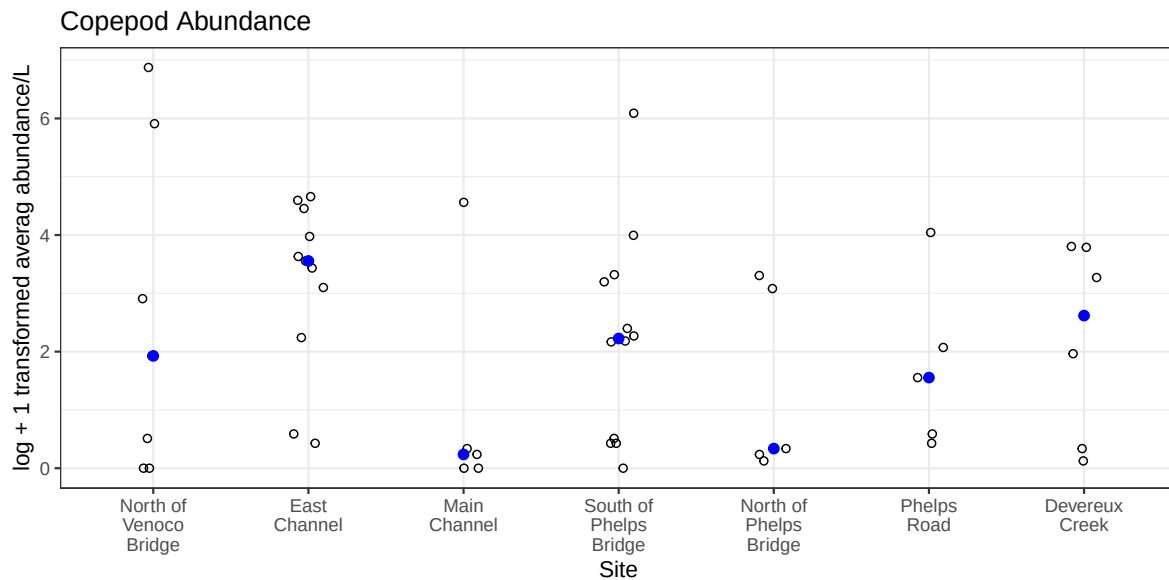
```

labs(x = "Site",
     y = "log + 1 transformed averag abundance/L",
     title = "Copepod Abundance") +

# different theme
theme_bw()

# display object
copepod_plot

```



1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site_full
- y-axis: average abundance of ostracod per liter
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
# creating new object from aquatic insects
ostracod <- aq_ins_clean |>

# filter to only include ostracod
filter(taxon == "ostracod") |>

# calculate the average of ostracod abundance per lit
mutate(ostracod_log_avg_per_lit = log(avg_per_lit + 1))

# display 10 random rows

slice_sample(ostracod, n = 10)
```

```
# A tibble: 10 x 24
  date_site date site taxon avg_per_lit med_ph med_sal med_do
  <chr> <dtm> <chr> <chr> <dbl> <dbl> <dbl> <dbl>
1 2022-11-12~ 2022-11-12 00:00:00 NEC ostr~ 0 7.33 7.01 2.4
2 2022-01-24~ 2022-01-24 00:00:00 NEC ostr~ 0.267 NA NA NA
3 2022-05-13~ 2022-05-13 00:00:00 NVBR ostr~ 0 9.72 74.2 11.4
4 2021-11-23~ 2021-11-23 00:00:00 NPB2 ostr~ 0.267 7.59 1.99 0.65
5 2021-04-15~ 2021-04-15 00:00:00 NPB ostr~ 185. 8.09 3.57 8.24
6 2021-09-01~ 2021-09-01 00:00:00 NPB2 ostr~ 0 NA NA NA
7 2022-05-06~ 2022-05-06 00:00:00 NMC ostr~ 0 9.96 1.45 16.5
8 2022-05-12~ 2022-05-12 00:00:00 NPB ostr~ 1.07 7.97 11.4 8.17
9 2023-05-05~ 2023-05-05 00:00:00 NPB ostr~ 24.7 NA NA 12.1
10 2022-08-12~ 2022-08-12 00:00:00 NEC ostr~ 0 7.37 NA NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, ostracod_log_avg_per_lit <dbl>
```

c. Code your figure

```
# base layer: create an object that stores the plot
ostracod_plot <- ggplot(data = ostracod,
  # x = site
  # y = log + 1 transformed average abundance/L
```



```

        mapping = aes(x = site_full,
                      y = ostracod_log_avg_per_lit)) +

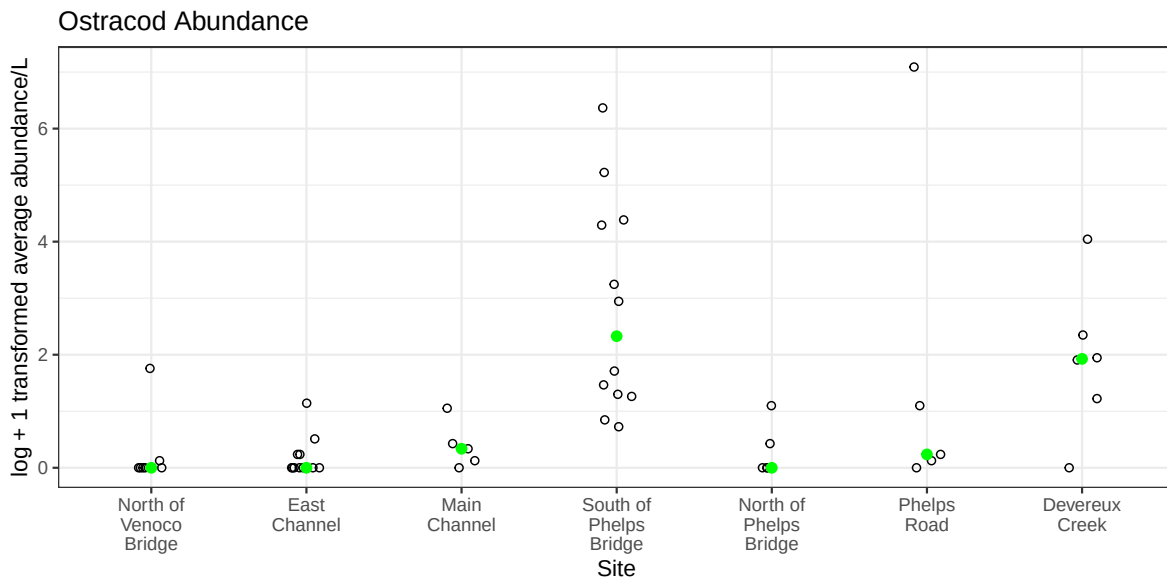
# first layer: geometry jitter points to represent each sample
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points to represent median abundance
stat_summary(geom = "point",
            fun = median,
            color = "green",
            size = 2) +
# wrapping x axis labels
scale_x_discrete(labels = label_wrap(width = 10)) +

# labeling axes and title
labs(title = "Ostracod Abundance",
     x = "Site",
     y = "log + 1 transformed average abundance/L") +

# changed theme
theme_bw()

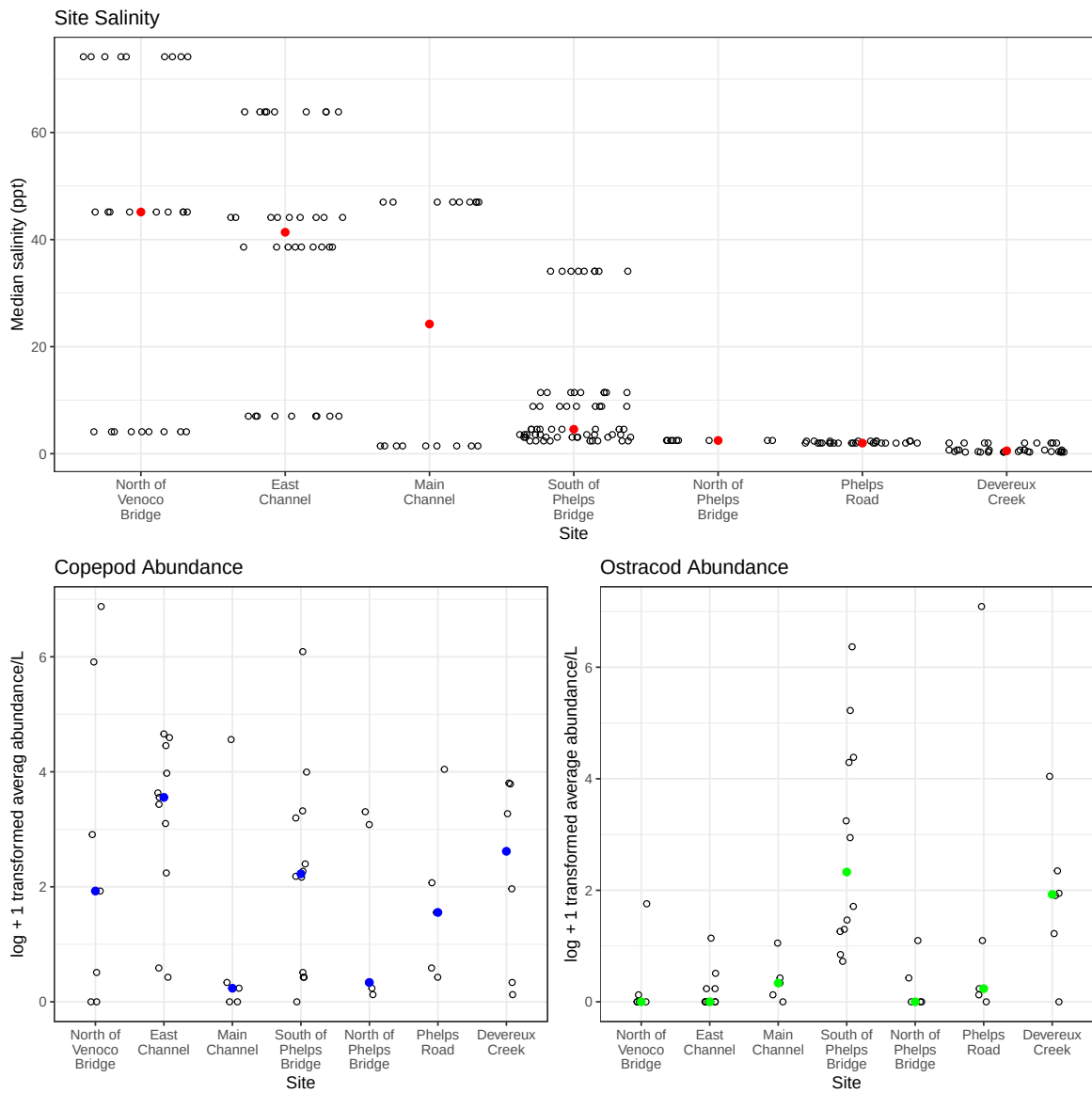
# display plot
ostracod_plot

```



d. Create a multipanel figure

```
salinity_plot / (copepod_plot | ostracod_plot)
```



e. Interpret your figure

What are the differences between sites in copepod and ostracod abundance?

- Average copepod abundance/L was highest at East Channel and Devereux Creek at respective transformed values of around 4 and 3, and moderate at a value of around 2 at North of Venoco Bridge, South of Phelps Bridge, and Phelps Road. The highest average ostracod abundance/L was moderate around a transformed value of 2 at South of Phelps Bridge and also Devereux Creek, and low at pretty much every other site.

How does ostracod abundance relate (or not relate) to site-level differences in salinity?

- Ostracod abundance does not correlate to salinity site-level differences in salinity because the plots do not show the same type of trend for the salinity plot and for the ostracod abundance plot.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: salinity (ppt)
- y-axis: log-transformed average abundance/L
- color: taxon
- geometry: geom_jitter()
- facets: by each taxon in aq_ins_clean with free scales

b. Wrangle your data

```
# change existing aq_in_clean object
aq_ins_clean <- aq_ins_clean |>

# change taxon values to title case
mutate(taxon = str_replace_all(taxon, "_", " ") |> str_to_title()) |>

# group by the sample date and by taxon
group_by(date, taxon) |>

# add in a new column that is the sum of avg_per_lit per taxon per date
mutate(total_avg_per_lit = sum(avg_per_lit, na.rm = TRUE)) |>

# log transform the total abundance per lit
mutate(log_total_avg_per_lit = log(total_avg_per_lit + 1)) |>
```

```
# ungroup
ungroup() |>

# keep only date, taxon, med_sal, and total avg/L
select(date, taxon, med_sal, log_total_avg_per_lit) |>

# re-order taxon levels by max abundance
mutate(taxon = fct_reorder(taxon, log_total_avg_per_lit, max, .desc =
  ↪ TRUE))

# display 10 random rows
slice_sample(aq_ins_clean, n = 10)
```

A tibble: 10 x 4

	date	taxon	med_sal	log_total_avg_per_lit
	<dtm>	<fct>	<dbl>	<dbl>
1	2021-04-15 00:00:00	Annelida Oligochaete	3.57	1.44
2	2021-11-12 00:00:00	Annelida Oligochaete	NA	0
3	2022-08-15 00:00:00	Annelida Polychaete	NA	0
4	2023-11-12 00:00:00	Annelida Oligochaete	47	0
5	2021-08-25 00:00:00	Hemiptera Corixidae Boatman	4.57	0
6	2020-02-21 00:00:00	Annelida Polychaete	4.1	0
7	2022-05-06 00:00:00	Hemiptera Corixidae Boatman	1.45	0
8	2022-01-24 00:00:00	Diptera Chironomid	NA	0
9	2022-02-25 00:00:00	Copepod	44.1	3.98
10	2021-11-16 00:00:00	Copepod	2.38	0.427

c. Code your figure

```
# base layer: create an object that stores the plot
abund_salinity_plot <- ggplot(data = aq_ins_clean,
  # x = salinity
  # y = log + 1 transformed average abundance/L
  mapping = aes(x = med_sal,
    y = log_total_avg_per_lit,
    color = taxon)) +

# first layer: geometry jitter points to represent each sample

geom_jitter(height = 0,
```

```

        width = 0.1,
        shape = 21) +

# facet by taxon
facet_wrap( ~taxon,
            scales = "free") +

# different color per taxon
scale_color_manual(values = natparks.pals("LakeNakuru", n = 9)) +

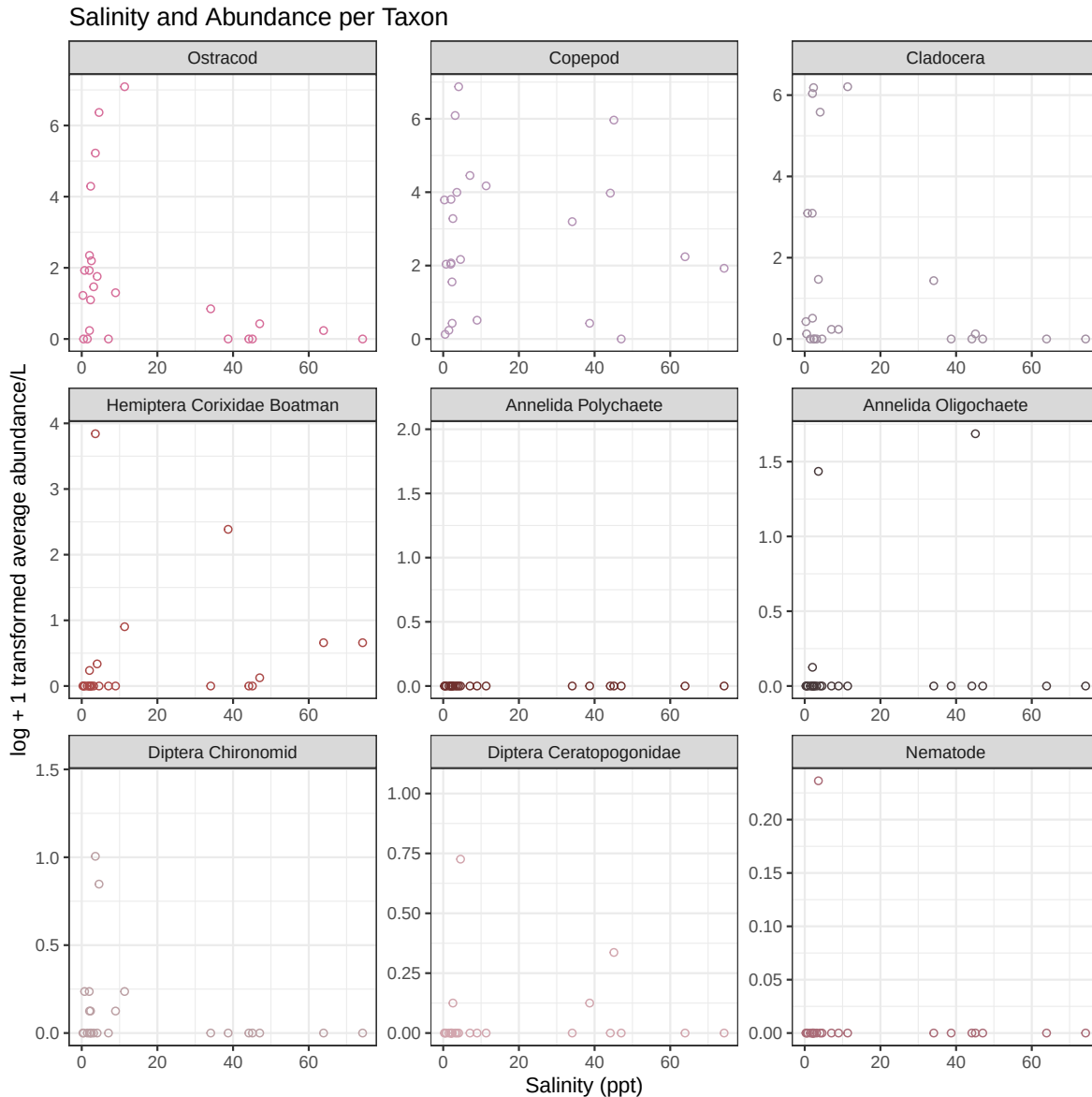
# labeling axes and title
labs(title = "Salinity and Abundance per Taxon",
      x = "Salinity (ppt)",
      y = "log + 1 transformed average abundance/L") +

# changed theme
theme_bw() +

# remove legend
theme(legend.position = "none")

# display plot
abund_salinity_plot

```



d. Interpret your figure

In 3-5 sentences, describe differences or similarities between taxa in relationships between abundance and salinity. Be specific using the visual components of the plot.

Across all sites, most of the data points are clustered in between a salinity of 0-10 ppt. Additionally, at all sites, there is a small cluster of data points around 40 ppt and 65 ppt, but not much in between 10-40 ppt and 50-60 ppt. For all taxa except for Annelida Polychaete and

Annelida Oligochaete, the highest abundance/L points lie in between this salinity range of 0-10 ppt. Ostracod, Copepod, and Cladocera have the most data points that range across average abundances per liter. Thus, it is reasonable to conclude that the lower salinity contributes to higher taxa abundance/L.